

Experiment 4 — Supervised Learning

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Aim

To perform correlation analysis on the Aadhar Enrollment dataset by computing and visualizing the relationships between enrollment variables across age groups using histograms, a Pearson correlation heatmap, and scatter plots, in order to identify the strength and direction of linear relationships among numeric features.

Theory

1. Feature Engineering: Raw datasets often do not contain all the variables needed for analysis. Feature engineering is the process of creating new meaningful columns from existing ones. In this experiment, a new column `Total_Enrollment` is derived by summing the three age-group columns — `age_0_5`, `age_5_17`, and `age_18_greater` — with missing values replaced by 0 using `fillna(0)` before summation. This gives a single unified enrollment count per record, which is useful for holistic analysis.

2. Distribution Analysis — Histograms: A histogram divides a continuous variable into equal-width intervals (bins) and plots the frequency of observations falling into each bin. It reveals:

- **Central tendency** — Where most values are concentrated (mean and median lines).
- **Spread** — How wide or narrow the distribution is.
- **Skewness** — Whether the distribution is symmetric, left-skewed (tail on left), or right-skewed (tail on right).

For enrollment data, histograms help identify whether most districts have low enrollment counts with a few outliers at the high end (right-skewed), which is common in demographic datasets.

3. Correlation Analysis — Pearson Coefficient: The Pearson Correlation Coefficient (r) measures the strength and direction of the linear relationship between two numeric variables. Its value ranges from -1 to $+1$:

Value	Interpretation
$r = +1$	Perfect positive linear relationship
$0 < r < 1$	Positive relationship (as one increases, so does the other)
$r = 0$	No linear relationship
$-1 < r < 0$	Negative relationship
$r = -1$	Perfect negative linear relationship

The correlation matrix is computed using `df.corr()`, which calculates pairwise Pearson coefficients for all numeric columns. Strong correlations between age groups would indicate that districts with high enrollment in one group tend to have high enrollment in others as well.

4. Correlation Heatmap: A heatmap is a 2D grid visualization where each cell's color encodes the magnitude and direction of the correlation coefficient between two variables. Darker or brighter colors represent stronger correlations. A heatmap makes it easy to quickly scan and compare all pairwise correlations at once, especially in datasets with many numeric features. Annotations inside each cell display the exact correlation value for precision.

5. Scatter Plots and Trend Lines: A scatter plot displays individual data points on a 2D plane, with one variable on the X-axis and another on the Y-axis. Each point represents one record. Scatter plots are used to:

- **Visually confirm correlations** — Points forming a clear upward or downward diagonal pattern indicate a linear relationship.
- **Detect outliers** — Points far from the main cluster stand out clearly.
- **Identify non-linear behavior** — If points form a curve rather than a line, a linear model may not be appropriate.

A trend line (regression line) is overlaid on each scatter plot to indicate the overall direction and slope of the relationship. In pairwise scatter plots across `age_0_5`, `age_5_17`, `age_18_greater`, and `Total_Enrollment`, the trend lines help visualize how strongly each pair of variables moves together.

6. Interpreting the Three Outputs Together: The three visualizations complement each other:

- **Histograms** reveal the individual distribution of each variable — skewness, concentration, and spread.
- **Heatmap** provides a quick numerical summary of all pairwise correlations in one view.
- **Scatter plots** visually confirm what the heatmap reports and expose outliers or anomalies that correlation numbers alone cannot reveal.

Together, they form a complete picture of how enrollment variables are distributed and how they relate to one another.

Execution

The following operations were performed step-by-step on the Aadhar Enrollment dataset:

Step 1 — Load and Inspect Data: The Excel file `Sample_Aadhar_Enrollment_Dataset.xlsx` was loaded into a Pandas DataFrame. Basic inspection was performed using `shape`, `info()`, `head()`, and `describe()` to confirm the dataset structure — 99 rows, 7 columns — and to identify data types and null counts.

Step 2 — Feature Engineering: A new column `Total_Enrollment` was created by summing `age_0_5`, `age_5_17`, and `age_18_greater` after filling missing values with 0. This column represents the total number of Aadhar enrollments per record regardless of age group.

Step 3 — Histogram Plotting: Histograms were plotted for each age group column and for `Total_Enrollment`. Mean and median lines were overlaid on each histogram to indicate central

tendency. The plots were saved as `aadhar_enrollment_histograms.png`. Each histogram was followed by a printed interpretation summary describing the skewness and concentration of enrollment values.

Step 4 — Correlation Matrix and Heatmap: The Pearson correlation matrix was computed using `df.corr()` on the numeric columns. A heatmap was generated using `seaborn.heatmap()` with annotations showing the exact correlation value in each cell. The heatmap was saved as `correlation_heatmap.png`. A printed interpretation summary followed, describing which pairs of variables had strong, moderate, or weak correlations.

Step 5 — Scatter Plots with Trend Lines: Pairwise scatter plots were generated for all combinations of `age_0_5`, `age_5_17`, `age_18_greater`, and `Total_Enrollment`. A linear trend line was fitted and overlaid on each scatter plot using `numpy.polyfit()` or `seaborn.regplot()`. The plots were saved as `aadhar_enrollment_scatter_plots.png`. A printed interpretation summary followed, describing the direction of each relationship and highlighting any visible outliers.

Summary of Outputs Generated:

Output File	Description
<code>aadhar_enrollment_histograms.png</code>	Age-wise and total enrollment frequency distributions
<code>correlation_heatmap.png</code>	Pearson correlation matrix visualized as a colored heatmap
<code>aadhar_enrollment_scatter_plots.png</code>	Pairwise scatter plots with trend lines showing variable relationships

Output

1) What It Does with the Aadhaar Dataset

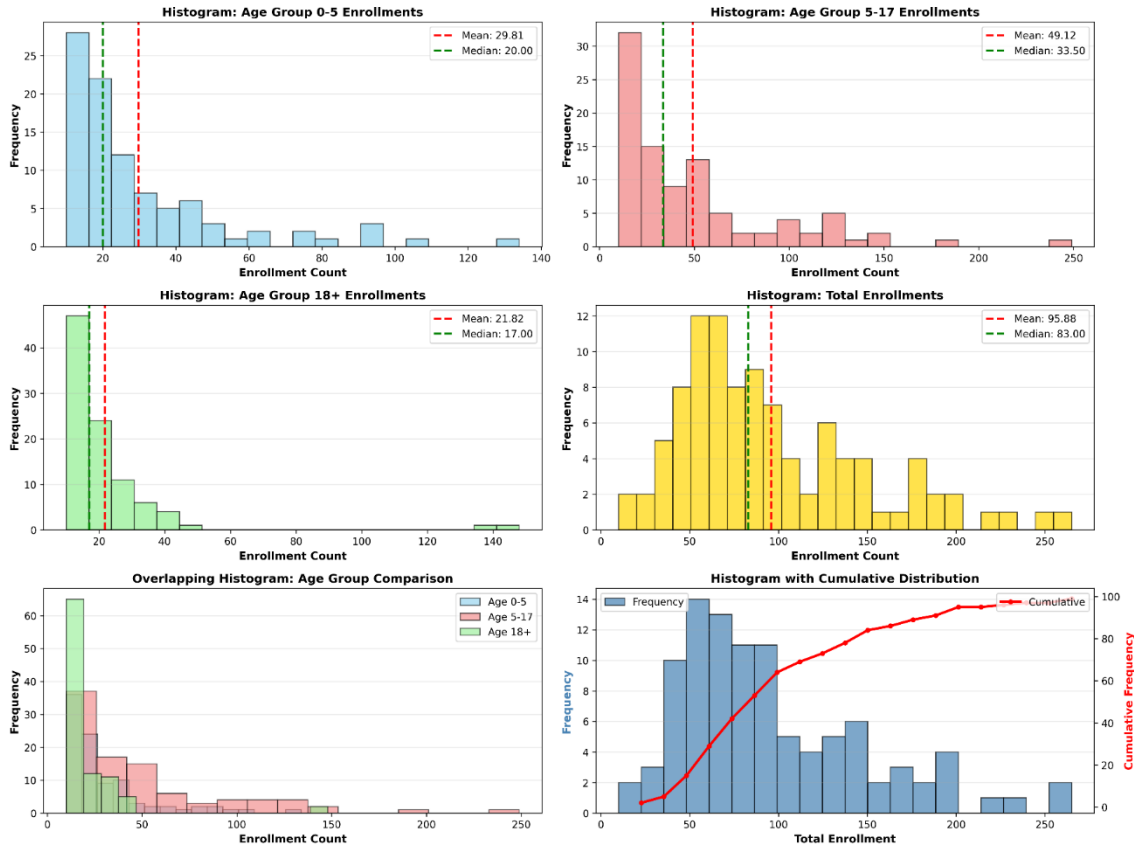
The program creates 6 histograms to analyze Aadhaar enrollment data for different age groups:

1. **Age 0–5 Histogram** – Shows enrollment distribution for children.
2. **Age 5–17 Histogram** – Shows school-age enrollments, which are the highest.
3. **Age 18+ Histogram** – Displays adult enrollment distribution.
4. **Total Enrollment Histogram** – Shows overall enrollment patterns.
5. **Overlapping Histogram** – Compares all age groups together.
6. **Cumulative Histogram** – Shows overall enrollment growth and percentiles.

Key Insights

- Most districts have lower enrollments, while a few have very high enrollments.
- The data is **right-skewed** because the mean is higher than the median.

- The 5–17 age group has the highest enrollment numbers.



GENERATING HISTOGRAMS

✓ Saved: aadhar_enrollment_histograms.png

HISTOGRAM ANALYSIS SUMMARY

Total Enrollments:
 - Mean: 95.88
 - Median: 83.00
 - Std Dev: 53.48
 - Range: 10 to 265

KEY INSIGHTS FROM HISTOGRAMS:

- The histograms show the frequency distribution of enrollments
- Age 5-17 group shows the highest enrollment counts on average
- Red dashed line indicates mean, green dashed line indicates median
- Most distributions show right-skewed patterns (positive skew)
- Cumulative histogram helps understand enrollment accumulation

HISTOGRAM GENERATION COMPLETE!

PS C:\Users\Asus\Desktop\TE\SEM 6\DS using Python LAB\EXP4>

2) What It Does with the Aadhaar Dataset

The heatmap analyzes the correlation between:

- Pincode
- Age 0–5
- Age 5–17
- Age 18+

- Total Enrollment

Color Meaning

-  **Red** → Strong positive correlation (both increase together)
-  **White** → Weak or no correlation
-  **Blue** → Negative correlation (one increases, other decreases)

Key Findings

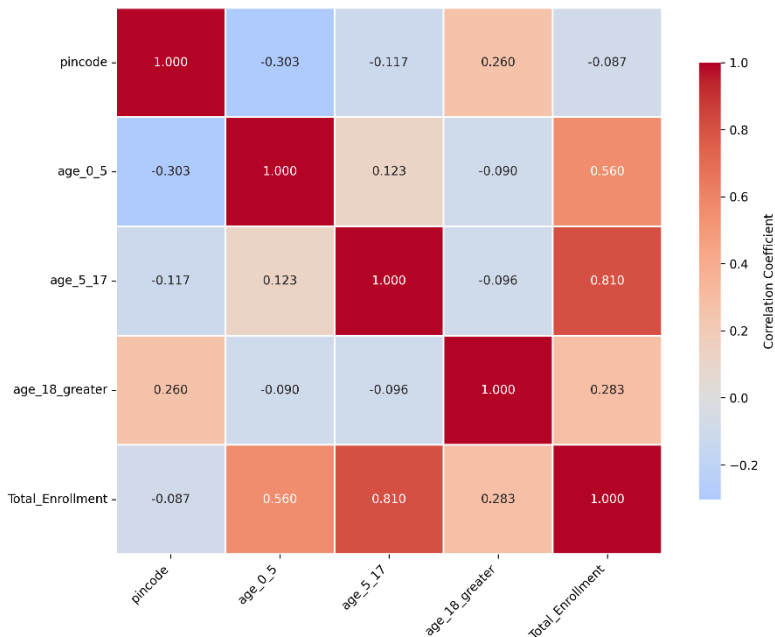
1. **Age 5–17 ↔ Total Enrollment = 0.810**
 - Strongest impact on total enrollments.
2. **Age 0–5 ↔ Total Enrollment = 0.560**
 - Moderate contribution to total enrollments.
3. **Pincode ↔ Age 0–5 = -0.303**
 - Weak negative relationship.
4. **Age 0–5 ↔ Age 18+ = -0.090**
 - Almost no relationship.

Practical Use

The heatmap helps identify:

- Which age group contributes most to enrollments
- Relationships between age groups
- Geographic trends using pincodes
- Important variables for predictions and analysis

Correlation Heatmap: Aadhar Enrollment Dataset



```
PS C:\Users\Asus\Desktop\TE\SEM 6\DS using Python LAB\EXP4> python EXP4.py
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CORRELATION ANALYSIS SUMMARY
=====

Correlation Matrix:
pincode    pincode  age_0_5  ...  age_18_greater  Total_Enrollment
pincode    1.000000 -0.303422 ...      0.260341      -0.086933
age_0_5    -0.303422 1.000000 ...      -0.090233      0.559962
age_5_17   -0.117133 0.122735 ...      -0.096018      0.810231
age_18_greater 0.260341 -0.090233 ...      1.000000      0.282533
Total_Enrollment -0.086933 0.559962 ...      0.282533      1.000000

[5 rows x 5 columns]

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KEY INSIGHTS FROM CORRELATION HEATMAP:
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• Positive correlation (red): Variables increase together
• Negative correlation (blue): One increases as other decreases
• Values range from -1 (perfect negative) to +1 (perfect positive)
• Values near 0 indicate weak or no linear relationship

Strongest Correlations:
• pincode ↔ age_0_5: -0.303
• age_0_5 ↔ Total_Enrollment: 0.560
• age_5_17 ↔ Total_Enrollment: 0.810
=====
```

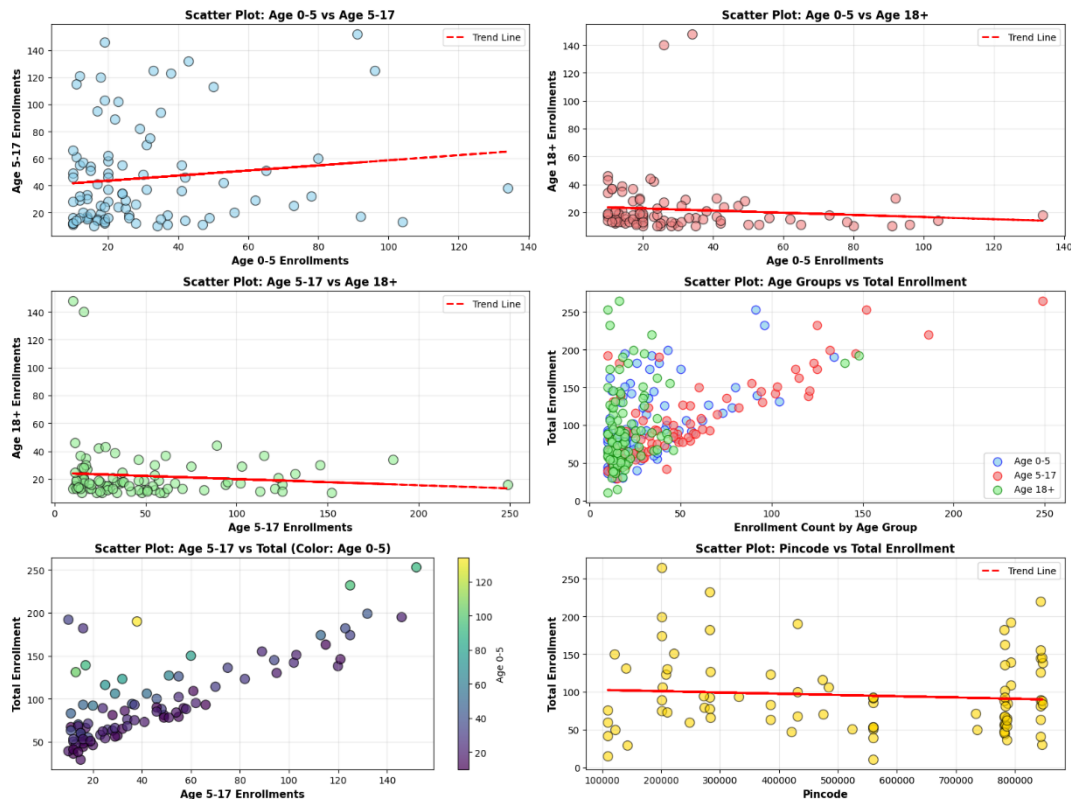
3) The analysis uses 6 scatter plots to study relationships between enrollment age groups.

1. **Age 0–5 vs Age 5–17**
 - Compares child and school-age enrollments.
 - Shows a weak positive relationship (0.123).
2. **Age 0–5 vs Age 18+**
 - Compares child and adult enrollments.

- Very weak negative relationship (-0.090).
- 3. **Age 5–17 vs Age 18+**
 - Compares school-age and adult enrollments.
 - Weak negative relationship (-0.096).
- 4. **Age Groups vs Total Enrollment**
 - Compares all age groups with total enrollments using different colors.
 - Age 5–17 has the strongest impact on total enrollments (0.810).
- 5. **Color Gradient Scatter Plot**
 - Shows Age 5–17 vs Total Enrollment with Age 0–5 shown using color intensity.
 - Helps visualize relationships between three variables together.
- 6. **Pincode vs Total Enrollment**
 - Checks whether location affects enrollments.
 - Very weak relationship (-0.087), meaning pincode has little effect.

Key Insight

The **5–17 age group** contributes the most to total Aadhaar enrollments, while other variables show weak relationships.



Conclusion

In this experiment, correlation and distribution analysis was performed on the Aadhaar Enrollment dataset. A new feature, **Total Enrollment**, was created by adding all three age-group enrollments. Histograms showed the distribution and skewness of enrollment data, while the correlation heatmap identified relationships between variables. Scatter plots with trend lines helped visualize these relationships and detect outliers.